

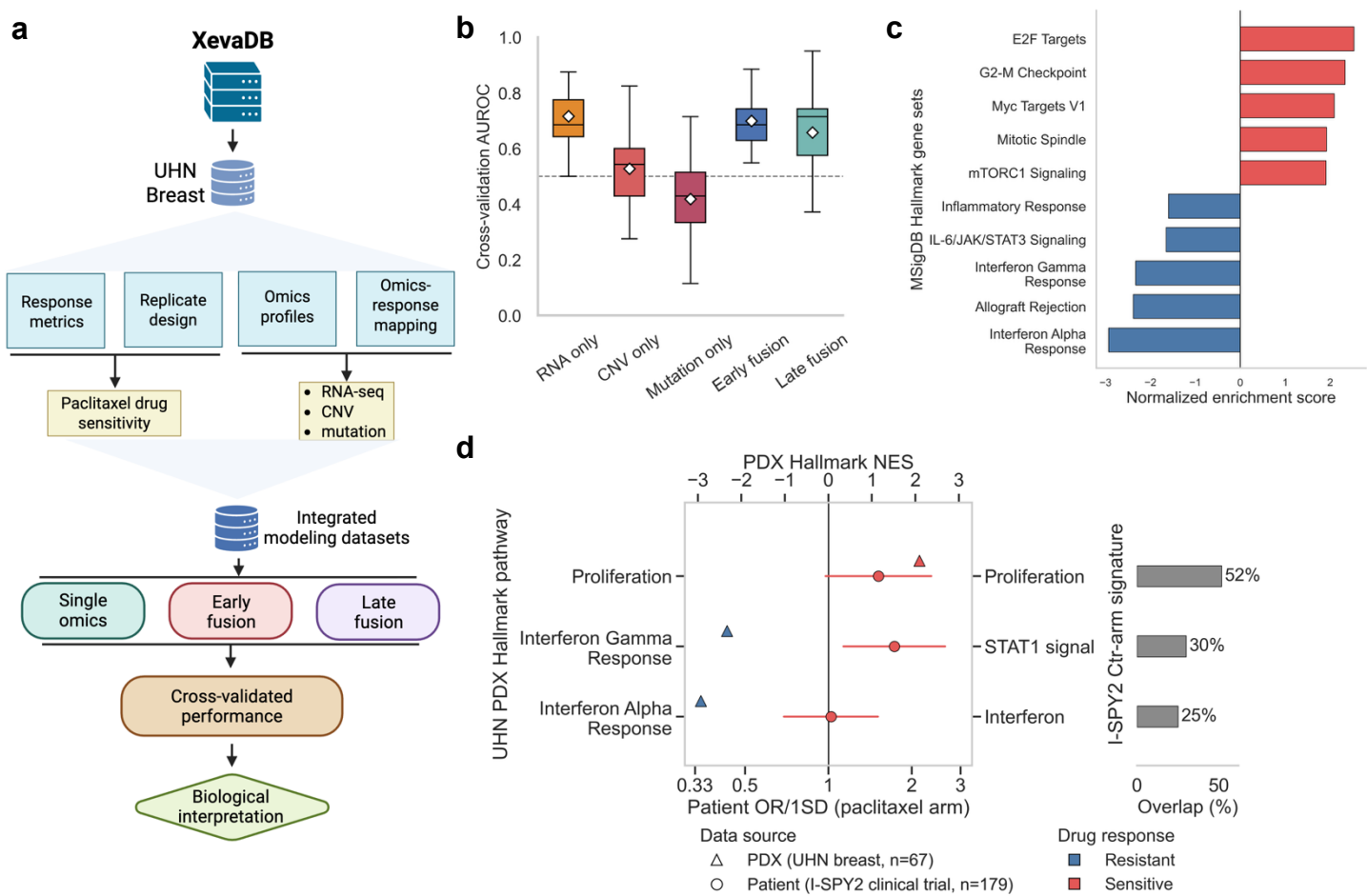


Figure 5. XevaDB-based multi-omics prediction of paclitaxel response in breast cancer PDX models.

(a) Case study four workflow: drug-sensitivity metrics and RNA-seq/CNV/mutation profiles for the UHN breast paclitaxel cohort are extracted from XevaDB and assembled into single-omics, early-fusion, and late-fusion modeling datasets, evaluated by cross-validation. (b) Cross-validation AUROC of the random forest classifier across single-omics (RNA-only, CNV-only, mutation-only) and multi-omics (early fusion, late fusion) feature sets; boxes show median/IQR and whiskers across repeated CV folds, diamonds mark the mean, dashed line marks chance (AUROC = 0.5). (c) Pre-ranked GSEA of the full RNA transcriptome against MSigDB Hallmark gene sets; top 6 significant pathways per direction shown (FDR $q < 0.05$). Positive NES: enriched in Sensitive models; negative NES: enriched in Resistant models. (d) Cross-cohort comparison of transcriptomic programmes associated with paclitaxel response in UHN PDXs ($n=67$) and pathological complete response (pCR) in the I-SPY2 paclitaxel control arm ($n=179$ patients). Triangles, PDX normalized enrichment scores (NES) for MSigDB Hallmark pathways; circles, patient odds ratios (OR) for pCR per one-standard-deviation (1SD) increase in the corresponding predefined I-SPY2 signature, with 95% confidence intervals (refit from patient-level I-SPY2 data; see Methods). Markers are colored by their own direction of association (blue, Resistant; red, Sensitive), so a row is concordant when both markers share a color and discordant when they differ. Positive PDX NES indicates enrichment in sensitive PDXs, whereas patient OR values above 1 indicate higher odds of pCR. "Proliferation" reflects the median NES across five Hallmark gene sets significantly enriched in sensitive PDXs (E2F Targets, G2-M Checkpoint, Mitotic Spindle, Myc Targets V1, Myc Targets V2). Percentages indicate gene overlap between each I-SPY2 signature and its corresponding Hallmark gene set. Not all pathways shown in (c) are evaluated here: mTORC1 Signaling, Allograft Rejection, IL-6/JAK/STAT3 Signaling and Inflammatory Response lacked an I-SPY2 signature with adequate gene-level overlap and are not shown.